ADVANCED STATISTICS IN PRACTICE

Implications of Advanced Statistics in Business and Research

Turning data into evidence, decisions, and discovery

PRESENTED BY **[Team Name]**

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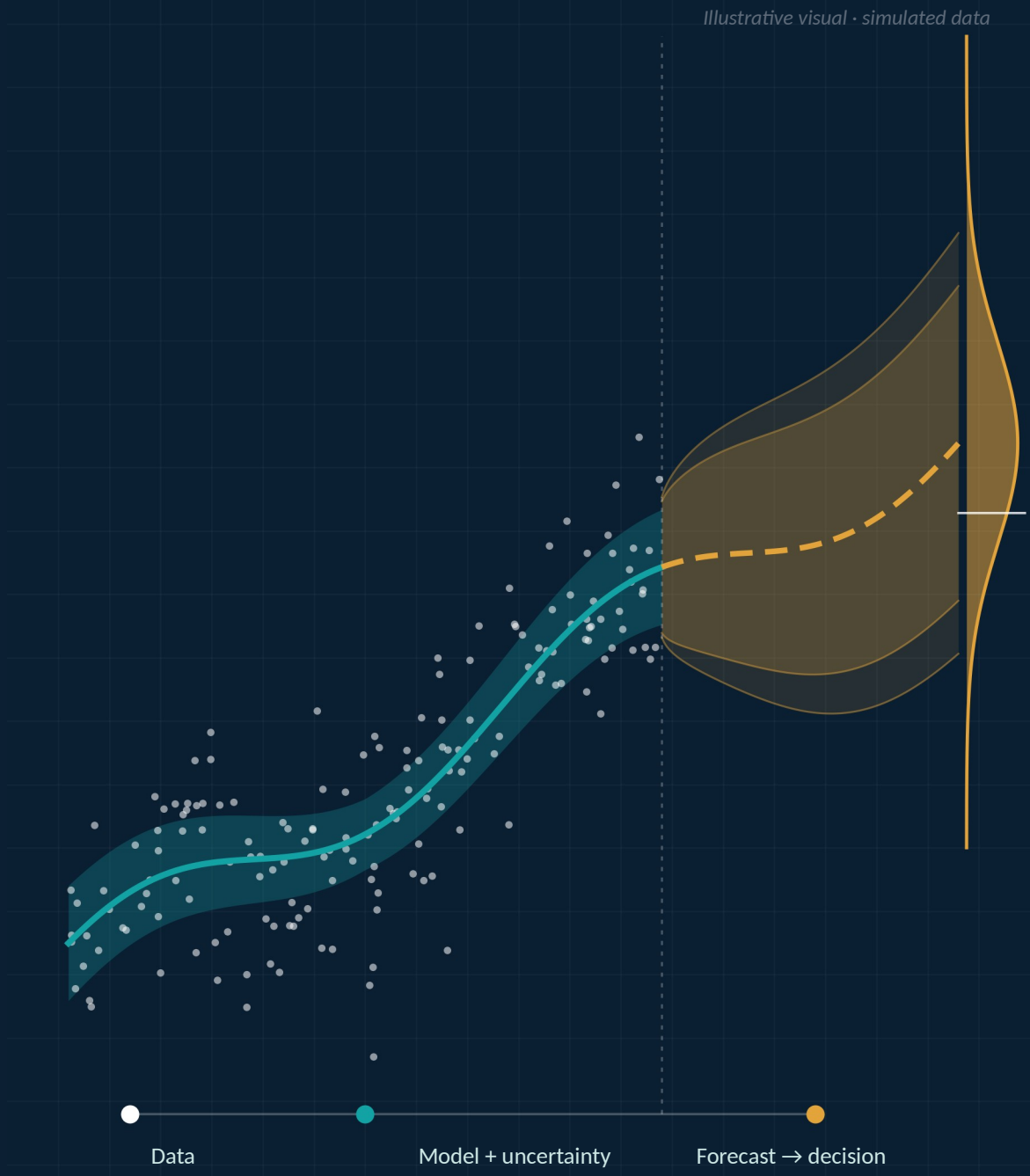
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[Instructor Name], [Designation]

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[Department]

UNIVERSITY
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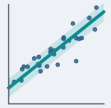
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From data to decisions

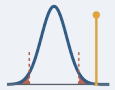
Raw data What was recorded? › **Description** What happened? › **Prediction** What is likely next? › **Inference** Is it real? How sure? › **Decision** What should we do?

Key methods: each answers a different question



Regression

How outcomes change with their drivers



Hypothesis tests & CIs

Is a difference real, and how big?



Bayesian inference

Update beliefs as new data arrives



Time-series forecasting

Predict the future, with a range



Experiments & causal inference

What happens if we act?



Clustering & multivariate

Find groups in complex data

EVIDENCE

5–6%

higher productivity in firms that use data-driven decision-making

Brynjolfsson, Hitt & Kim (2011) · observational study of 179 large firms

Better decisions in business and research

Business

- Demand forecasting
- Customer segments
- Pricing
- Credit risk & fraud

REAL CASE · BING SEARCH ADS

IDEA	METHOD	RESULT
Show longer ad titles	Randomized A/B test	**+12% revenue** ≈ $100M a year in the U.S.

Kohavi & Thomke (2017), Harvard Business Review

Research

- Study design
- Power analysis
- Confidence intervals
- Reproducibility

REAL CASE · RECOVERY TRIAL

28-day deaths in ventilated COVID-19 patients



Randomized trial · RECOVERY Collaborative Group (2021), NEJM

ILLUSTRATIVE DATA hypothetical, not real company results

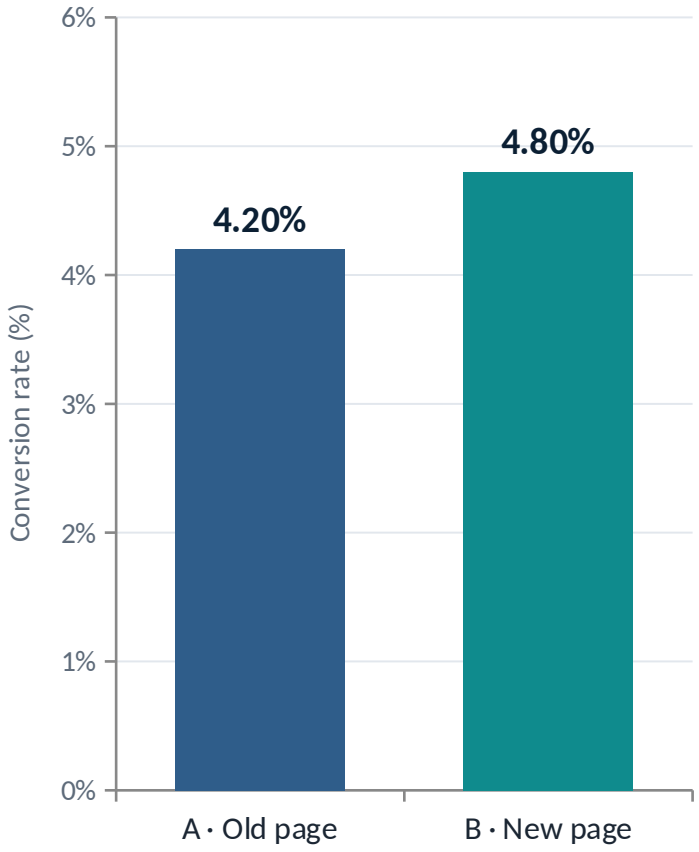
A/B test: does a new product page sell more?

1 Split
40,000 visitors at random: 20,000 see the old page (A), 20,000 the new page (B).

2 Measure
Conversion rate = orders ÷ visitors.

3 Test
Two-proportion z-test at the 5% level.

Visitors are assigned at random, so the only real difference between the groups is the page.



A: 840 / 20,000 · B: 960 / 20,000 orders

+0.60 pp
difference (B − A)

+0.19 to +1.01 pp
95% confidence interval

p = 0.004
very unlikely to be chance

Meaning The new page very likely works better, but the exact gain is uncertain.

DECISION
Launch page B and keep monitoring results.

Use statistics carefully and honestly

Biased data

Wrong or one-sided data gives wrong answers.

Overfitting

A model that memorizes noise fails on new data.

P-hacking

Running many tests guarantees some false “wins”.

Correlation ≠ causation

Even accurate AI predictions do not prove cause.

Significant ≠ important

Always report how big the effect is.

Privacy & fairness

Protect people and check models for bias.

RESPONSIBLE PRACTICE

1 Plan › 2 Check › 3 Estimate › 4 Stress-test › 5 Report

set the question first — data and assumptions — effects with uncertainty — robustness checks — honestly, with limits

05 · CONCLUSION

Key takeaways

01

Data becomes evidence

Statistics separates real patterns from noise.

02

Test before you decide

Experiments and intervals beat opinion.

03

Design drives rigor

Good design and sample size make research reliable.

04

Interpret responsibly

Effect size, transparency and fairness matter.

“The value of data lies not merely in collecting it, but in drawing reliable conclusions from it.”

Thank you · Questions?

References: Brynjolfsson, Hitt & Kim (2011), SSRN · Kohavi & Thomke (2017), Harvard Business Review 95(5) · RECOVERY Collaborative Group (2021), NEJM 384(8) · Open Science Collaboration (2015), Science 349 · Wasserstein & Lazar (2016), The American Statistician 70(2)